

# Predictive QoS Routing for Connected Vehicles via Physics-Informed Graph Neural Networks

## Executive Summary

Recent work has begun to combine radio-propagation physics with machine learning and graph neural networks (GNNs) to build radio environment maps (REMs), optimize mobile networks, and enable predictive QoS for vehicular applications. However, existing studies typically (i) focus on predicting generic signal strength/REM rather than per-road-segment vehicular throughput, (ii) use simplified or implicit environment representations (e.g., grid maps, aerial images, ray-tracing graphs) without explicit sky-view factor (SVF) and terrain shadowing, (iii) decouple coverage prediction from actual routing decisions, and (iv) do not integrate street-level computer vision (CV) for foliage attenuation or online map-healing from vehicle telemetry.<sup>[1][2][3][4][5][6][7][8][9]</sup>

The proposed architecture is novel primarily in its integration of a deterministic physics engine (OSM/3D buildings + OpenCellID towers + SRTM terrain with SVF, Okumura-Hata, and 3D shadowing), a road-graph GNN calibrator trained on sparse Ookla throughput tiles, and an edge feedback loop using V2C telemetry and SegFormer-based tree-canopy detection to support predictive QoS routing for connected vehicles. The individual components (Okumura-Hata, GNNs for REMs, foliage loss models, predictive QoS concepts) are present in the literature, but their fusion in a single end-to-end automotive routing stack appears to be missing, creating a clear publishable novelty gap.<sup>[5][6][10][11]</sup>

## 1. Literature Overlap

### 1.1. GNN-based radio map and propagation modeling

#### 1.1.1 Data-Driven Radio Environment Map Estimation Using GNNs

Shibli and Zanoluda (2024) propose a GNN-based method to estimate radio environment maps using sparse geo-located RSRP/RSRQ measurements together with physical cell information. The approach partitions the coverage area into a graph, encodes spatial proximity and cell attributes as node/edge features, and trains a GNN to interpolate signal strength over unmeasured locations, effectively learning a data-driven propagation surrogate.<sup>[^5]</sup>

What is missing relative to our architecture:

- The graph is defined over generic spatial tiles (cells of a REM) rather than a road network; there is no notion of per-segment vehicular QoS or routing decisions on top of the predictions.<sup>[^5]</sup>
- The environment representation is abstract (tiles plus cell metadata); it does not incorporate explicit SVF via 360° ray-casting, 3D terrain shadowing, or a deterministic

Okumura-Hata path-loss backbone derived from OSM buildings and SRTM.<sup>[12][5]</sup>

- The learning target is signal strength (RSRP/RSRQ), not application-level throughput calibrated against crowdsourced speed tests such as Ookla; vehicular context and V2C feedback are outside the scope.<sup>[11][5]</sup>

### 1.1.2 GNN-based Radio Map Construction in Urban Environments

Chen et al. (2023) present a GNN-based radio map construction method for urban environments in IEEE Communications Letters. They recover a complete urban radio map from sparse measurements by representing sampling locations as nodes in a graph and learning spatial dependencies via message passing; the method outperforms classical interpolation techniques in accuracy and robustness.<sup>[^7]</sup>

Gaps relative to our work:

- The graph is over measurement locations, not over a transportation network; radio-map estimation is the end goal, with no explicit coupling to vehicle routing or V2X QoS guarantees.<sup>[^7]</sup>
- The environmental context appears through distance and possibly building-related features, but there is no explicit SVF computation, SRTM-based terrain modeling, or use of Okumura-Hata as a physics prior.<sup>[12][7]</sup>
- No integration with crowdsourced throughput datasets such as Ookla tiles, no online map-healing from vehicular telemetry, and no CV-based foliage modeling are reported.<sup>[11][7]</sup>

### 1.1.3 Data-Driven Radio Propagation Modeling Using GNNs (Orange Labs)

Bufort et al. (Orange Labs, 2023/2025) propose a data-driven radio propagation modeling framework using GNNs. They convert the radio environment into a graph where nodes represent locations and edges encode both spatial adjacency and ray-tracing relationships; a GNN is trained on real measurement data to predict propagation characteristics, aiming to replace inflexible analytic models.<sup>[4][13]</sup>

Overlap and differences:

- This is close in spirit to using a physics-motivated graph structure (ray-tracing relationships) as input to a GNN, but the representation is still generic spatial nodes rather than road segments explicitly tailored to vehicular movement.<sup>[^4]</sup>
- While ray-tracing relationships appear, there is no explicit SVF scalar feature nor a full 3D building-terrain fusion from OSM + SRTM as in the proposed deterministic engine.<sup>[14][4]</sup>
- The work focuses on propagation prediction; it does not aim to deliver a predictive QoS routing engine, nor does it integrate online V2C telemetry or CV-based foliage attenuation.

### 1.1.4 Data-Enhanced GNN with Simplex Vectors for Radio Map Prediction

A recent IEEE study on data-enhanced GNNs with simplex vector augmentation for radio map prediction proposes to recover RSS maps from partial random RSS samples using a GNN that operates over observed points. The method leverages simplex-based feature embeddings to better capture spatial structure and improve interpolation performance.<sup>[15]</sup>

Missing elements relative to our system:

- Although GNNs are used for REM construction, the approach again targets generic spatial interpolation rather than mapping QoS onto road segments for vehicle routing.<sup>[15]</sup>
- The physics side is relatively light: there is no explicit Okumura-Hata modeling, SVF, or terrain shadowing as engineered features intertwined with the GNN.<sup>[15][12]</sup>
- No edge-case handling via V2C telemetry feedback or CV-based vegetation mapping is considered.

## 1.2. Machine-learning-enhanced path-loss and Okumura-Hata models

### 1.2.1 Surveys and ML-based Path-Loss Prediction

Erunkulu et al. (2020) provide a survey of cellular coverage prediction techniques, emphasizing that many propagation models are extensions of Okumura-Hata and discussing their frequency ranges, terrain suitability, and comparative performance. More recently, Halifa et al. (2025) survey machine-learning-based path-loss models, reviewing shallow and deep learning approaches (e.g., random forests, CNNs, hybrid models) that often use classical models such as COST-231/Okumura-Hata as baselines or features.<sup>[16][10][8]</sup>

Neural-network-based path-loss prediction methods demonstrate that ML can substantially reduce RMSE compared to pure empirical models by learning corrections from measurement data. Several works generate synthetic training sets via ray tracing and then validate ML models (e.g., random forest, KNN) on real measurements, particularly in sub-6 GHz suburban environments.<sup>[17][18][19][8]</sup>

What they do not do:

- These studies primarily treat Okumura-Hata (and variants) as either baselines or scalar features; they rarely embed a rich 3D city/terrain model with SVF and ray-casting into a path-loss feature vector for each road segment.<sup>[10][16][17]</sup>
- They predict path loss or RSS, not end-to-end application throughput along routes for moving vehicles, nor do they build a graph-structured calibrator aligned with navigation networks.
- Integration with crowdsourced consumer speed tests (e.g., Ookla Open Data tiles) as supervisory signals is not mainstream, even though Ookla makes such tiles publicly available.<sup>[20][11]</sup>

## 1.2.2 Optimization and Hybridization of Hata-type Models

Some works explicitly optimize the Hata/Okumura-Hata equations using stochastic optimization or regression techniques to better fit local measurement campaigns. Others enhance path-loss simulators (e.g., for LoRa or sub-GHz systems) with ML-based correction layers trained from mobile measurements in urban scenarios.<sup>[21][22][19]</sup>

Compared to our architecture:

- These efforts improve scalar path-loss formulas but generally lack graph-structured learning, road-network awareness, or integration into vehicular QoS routing engines.<sup>[19][21]</sup>
- Vegetation and building attenuation are handled via standard empirical models (e.g., ITU-R P.833, COST235), not via CV-derived, street-level canopy features fused with propagation physics.<sup>[23][9]</sup>

## 1.3. Predictive QoS and routing for vehicular networks

### 1.3.1 Route Planning with 5G QoS Prediction

Liu et al. (2025) propose a route-planning scheme with 5G QoS prediction based on probability distribution detection (PDD) to improve QoE for V2X services. They combine two QoS prediction models and compute probability confidence weights to select routes that maximize expected throughput, reporting measurable gains over shortest-path routing in simulations.<sup>[1]</sup>

Limitations relative to our pipeline:

- Their QoS prediction is based on statistical models at the network level; the paper does not describe a physics-informed geospatial engine with SVF, Okumura-Hata, or detailed 3D building and terrain data.<sup>[1]</sup>
- The learning architecture appears to fuse multiple predictors at the scalar QoS level; it does not use a GraphSAGE-type GNN over a road graph with rich physics features per segment.<sup>[1]</sup>
- There is no mention of online map-healing from vehicle telemetry or CV-based vegetation attenuation; environment dynamics (e.g., growing trees, new buildings) are not explicitly addressed.

### 1.3.2 Multi-MNO Predictive-QoS for Vehicular Applications

Mittal and Leinmüller (2021) introduce a concept of predictive-QoS (P-QoS) for vehicular applications across multiple mobile network operators (MNOs). They propose an entity that consumes QoS predictions for different MNOs over a region and plans MNO usage for a vehicle (e.g., operator selection, time-plans) to maintain service quality during roaming.<sup>[2]</sup>

Gaps versus our work:

- P-QoS is treated at the operator-selection layer; the paper does not specify how QoS predictions are generated from physical environment data or ML models.<sup>[24][2]</sup>
- There is no explicit road-segment-level prediction tied to routing decisions; QoS time-series are considered for areas rather than for fine-grained route segments with building/terrain context.<sup>[^2]</sup>
- CV-based foliage modeling, SVF computation, and deterministic path-loss engines are outside the scope.

### 1.3.3 QoS Prediction for 5G Connected and Automated Driving / 5GAA Reports

A 2021 arXiv article describes how 5G systems can generate predictive QoS information (e.g., predicted throughput, latency) and deliver it to vehicles to anticipate QoS changes for connected and automated driving. The 5GAA technical report on predictive QoS and V2X service adaptation further specifies interfaces whereby networks expose probabilistic QoS forecasts to V2X applications.<sup>[3][24]</sup>

Relative to our architecture:

- These works formalize predictive QoS as a network service but are largely agnostic to the specific propagation modeling and ML pipeline; they do not detail physics-informed GNN architectures or map-healing loops.<sup>[3][24]</sup>
- The focus is on signaling and adaptation mechanisms; route optimization itself and geometric modeling of the environment (buildings, terrain, trees) are abstracted away.

## 1.4. Environment and foliage modeling with CV and remote sensing

There is extensive literature on SVF computation and its application in urban climates and radiation modeling, typically using LiDAR or rasterized height data to compute SVF via ray-casting across multiple directions. However, these works target thermal and radiative processes rather than cellular QoS, and do not fuse SVF with telecom path-loss models.<sup>[25][26][^14]</sup>

Vegetation attenuation has been studied in detail for 5G/mmWave and sub-GHz systems using empirical models (e.g., ITU-R P.833, Weissberger, COST235, FITU-R) and measurement campaigns. These models treat foliage depth and frequency as key predictors of excess loss but are not commonly linked to computer-vision-derived canopy maps.<sup>[27][9][28][23]</sup>

On the CV side, multiple works use deep segmentation models (sometimes transformer-based) to detect urban tree canopy or building structure from aerial or street-level imagery for ecological or urban-planning objectives. SegFormer-like architectures are widely used in autonomous-driving perception pipelines, but there is little evidence of such models being used to derive foliage attenuation penalties for cellular propagation maps.<sup>[29][30][31][32]</sup>

In summary, while each ingredient of the proposed system has an analogue in existing literature (SVF computation, vegetation attenuation models, CV-based canopy mapping),

there is no integrated pipeline that uses street-level segmentation (e.g., SegFormer) to adjust attenuation in a physics-informed REM used for vehicular QoS routing.

## 2. The Novelty Gap

### 2.1. Physics-informed GNN over a road network

The closest works on GNN-based radio map estimation and propagation modeling represent the environment as generic spatial graphs or ray-tracing graphs, trained to interpolate signal strength from sparse measurements. In contrast, the proposed architecture builds a graph directly on the road network (e.g., from OSM), treating road segments or intersections as nodes, and encodes 13 engineered physics features per segment derived from SVF, Okumura-Hata path loss, building blockage, and 3D terrain shadowing.<sup>[14][4][12][7][5][15]</sup>

This represents a tighter coupling between propagation physics and the graph structure relevant for vehicular routing, rather than merely learning a generic REM that must be post-processed for routing. None of the surveyed GNN-REM works explicitly combine a deterministic path-loss backbone with SVF and terrain features as inputs to a road-graph GNN calibrated on crowdsourced throughput.<sup>[4][7][5][15]</sup>

### 2.2. Crowdsourced throughput calibration on top of Okumura-Hata

Okumura-Hata and its variants are standard and widely used; multiple surveys and case studies confirm both their ubiquity and their limitations in heterogeneous 4G/5G environments. From a pure propagation-modeling standpoint, using Okumura-Hata as the deterministic backbone is not novel.<sup>[16][10][^21]</sup>

The novelty lies instead in how the model is used:

- Okumura-Hata features (augmented with SVF and 3D shadowing) are fed into a GraphSAGE GNN that learns corrections against sparse, real-world throughput from a crowdsourced dataset (e.g., Ookla Open Data tiles or similar), effectively implementing a physics-guided surrogate for application-level bandwidth.<sup>[10][11][^5]</sup>
- The calibration is performed on a road-aligned graph, so that unmeasured road segments inherit both physics priors and data-driven corrections via message passing; this is different from the tile-based REM interpolation in existing work.

To the best of the surveyed literature, there is no work that explicitly fuses an Okumura-Hata-based deterministic engine, SVF, 3D terrain, and crowdsourced throughput in a GNN over the road network for predictive vehicular QoS.

### 2.3. Integration of CV-derived canopy attenuation into a live REM

Vegetation attenuation has rich empirical models and multiple measurement-based studies across frequencies, but these rely on manually specified foliage depths or uniform canopy assumptions. In parallel, SegFormer and related architectures are state-of-the-art for semantic segmentation of street-level imagery (buildings, vegetation, road, etc.), and there is

a growing body of work on mapping urban canopy using point clouds and multimodal imagery.<sup>[30][9][28][31][32][33][27][23][^29]</sup>

What appears novel is to close the loop between these domains:

- Using SegFormer (or similar) on street-level imagery along roads to derive fine-grained canopy masks or volumetric estimates for vegetation directly above and beside the road.<sup>[31][29]</sup>
- Converting those masks into additional attenuation penalties that augment or override standard foliage models, thereby adjusting per-segment path loss in the REM in a data-driven fashion.
- Running this as part of an online V2C telemetry loop, where mispredictions in bandwidth trigger targeted CV reanalysis or parameter updates to correct the map locally.

Existing work does not report an end-to-end system where CV-derived canopy features dynamically adjust propagation maps used for vehicular QoS prediction and routing.

## 2.4. End-to-end predictive QoS routing for connected vehicles

Predictive QoS as a network service and its integration with V2X applications has been conceptually defined in both academic and industry-driven work (e.g., 5GAA), and there are initial demonstrations of QoS-aware route planning in simulated 5G networks. However, these approaches typically assume that QoS predictions (e.g., mean throughput, outage probability) are already available per link or area, without detailing how they are generated from physical environment data and crowdsourced measurements.<sup>[34][24][2][3][^1]</sup>

The proposed architecture fills this implementation gap by:

- Generating per-road-segment bandwidth predictions from a physics-informed, GNN-calibrated REM, rather than treating QoS as an exogenous time-series.<sup>[5][1]</sup>
- Directly integrating these predictions into routing algorithms (e.g., multi-objective shortest path maximizing minimum expected bandwidth or minimizing probability of outage), thus closing the loop from physical environment to route choice.
- Incorporating an online map-healing mechanism via V2C telemetry, allowing the REM and the GNN calibration to adapt over time as networks evolve.

This end-to-end chain—from open geodata and crowdsourced throughput, through physics-informed GNN calibration, to vehicular route optimization—does not appear in the reviewed literature.

## 2.5. Critical assessment of novelty and standard components

From a peer-review perspective, several components should be explicitly framed as standard to avoid overstating novelty:

- The use of Okumura-Hata as the baseline path-loss model is conventional; numerous studies have applied it to 4G/5G planning and compared it against newer empirical models.<sup>[35][21][16][12]</sup>

- GraphSAGE with max aggregation is a mainstream GNN architecture; similar message-passing models have already been applied to radio map construction and mobility relation modeling in operational networks.<sup>[7][4][^5]</sup>
- SegFormer itself is a standard segmentation backbone for street-scene understanding; using it for tree detection is an application of known CV tooling.<sup>[^36]</sup>

The publishable novelty resides in the system-level integration and in the specific fusion of deterministic physics, crowdsourced throughput, GNN-based spatial calibration over a road graph, and CV-based vegetation modeling for predictive QoS-aware routing.

### 3. Paper Angles and Framing

#### 3.1. Angle 1 – Telecommunications / RF physics focus

Proposed title:

Physics-Informed Graph Neural Networks for Radio Environment Mapping and Predictive QoS in Urban Vehicular Scenarios

Key framing elements:

- Emphasize the deterministic physics engine: fusion of OSM buildings, OpenCellID tower metadata, and SRTM terrain; computation of SVF via 360° ray-casting; use of Okumura-Hata (or COST-231 Hata where appropriate) as the baseline large-scale path-loss model; and explicit modeling of 3D terrain shadowing.<sup>[16][14][^12]</sup>
- Position the GNN primarily as a data-driven correction layer on top of a carefully engineered physics feature space, highlighting improvements in prediction accuracy over pure Okumura-Hata and pure data-driven models.<sup>[17][10][^5]</sup>
- Present detailed ablation studies showing the incremental contribution of SVF, terrain shadowing, and canopy attenuation compared to standard clutter-based models (e.g., ITU-R, COST235) in urban canyons and tree-lined streets.<sup>[9][27][^23]</sup>
- For routing, focus on how improved large-scale fading modeling translates into better predictive QoS estimates (e.g., reduced outage probability along routes), rather than on automotive systems per se.

This angle will appeal to RF propagation and network planning audiences by demonstrating that physics-informed ML provides a practical compromise between full ray tracing and purely empirical REM interpolation.

#### 3.2. Angle 2 – AI / GNN architecture focus

Proposed title:

Road-Graph Neural Networks for Physics-Guided QoS Prediction from Crowdsourced Cellular Measurements

Key framing elements:

- Highlight the construction of a heterogeneous road graph where nodes/edges are annotated with a rich set of physics features (SVF, tower distance and frequency, Okumura-Hata loss, terrain blockage, canopy attenuation) and targets are derived from aggregated Ookla throughput tiles.<sup>[10][11][^5]</sup>
- Describe the GraphSAGE-based model with max aggregation, explain why this architecture is appropriate for sparse, irregular measurement patterns, and compare it against grid-based CNNs, standard MLPs, and alternative GNNs (e.g., GAT, GraphConv) for spatial generalization.<sup>[4][15][7][5]</sup>
- Emphasize the physics-guidance aspect: show that including deterministic features improves sample efficiency, generalization to unseen roads, and robustness to distribution shift (e.g., after network upgrades) compared to purely data-driven baselines.
- Discuss the online learning and map-healing loop as a continual-learning problem where V2C telemetry provides new labeled edges/nodes and show how the model can be incrementally updated without catastrophic forgetting.

This angle fits well into AI for telecom and graph learning venues, positioning the contribution as a novel graph formulation and learning paradigm for QoS prediction.

### 3.3. Angle 3 – Automotive / ITS application focus

Proposed title:

Predictive QoS Routing for Connected Vehicles via Physics-Calibrated Road-Graph Neural Networks

Key framing elements:

- Frame the work as an ITS contribution solving the gap identified by predictive QoS initiatives (e.g., 5GAA) by providing a concrete implementation that maps environmental data and crowdsourced throughput into actionable route choices.<sup>[24][34][2][3]</sup>
- Emphasize vehicular QoE and safety-critical use cases (e.g., tele-operated driving, cooperative perception) that require not just instantaneous coverage but forecasted throughput along candidate routes.<sup>[34][3]</sup>
- Compare traditional shortest-path or travel-time-optimal routing against the proposed QoS-aware routing under various scenarios (urban canyon, suburban tree-lined roads, rural terrain shadowing), showing reductions in expected outage time or improvements in minimum guaranteed bandwidth.
- Highlight the integration of in-vehicle sensing (street-level imagery for canopy detection, V2C telemetry for QoS feedback) with cloud-based REM and GNN calibration as a concrete example of vehicle-to-cloud intelligence.

This framing will resonate with ITS and automotive networking communities by demonstrating tangible mobility benefits rather than focusing on propagation modeling details.

## 4. Target Journals and Conferences

### 4.1. Telecommunications / RF and network optimization venues

- **IEEE Transactions on Vehicular Technology (TVT):** Publishes work at the intersection of wireless communications and transportation; prior issues include radio-propagation modeling for vehicular channels, REM construction, and predictive QoS for V2X, making it a good fit for the physics-informed GNN plus routing angle.<sup>[2][1][^5]</sup>
- **IEEE Transactions on Wireless Communications (TWC) or IEEE Transactions on Communications (TCOM):** Suitable if the focus is primarily on propagation modeling and GNN-based REM estimation; several recent papers on ML for radio propagation and REMs have appeared in these venues.<sup>[8][4][^5]</sup>
- **IEEE ICC / IEEE GLOBECOM (symposia on Wireless Communications, Network Softwarization, or AI for Networks):** Good options for a first conference publication emphasizing the system architecture and empirical gains of physics-guided GNNs for QoS prediction.<sup>[15][4][^5]</sup>

### 4.2. AI for telecom and graph learning venues

- **IEEE INFOCOM or IEEE/ACM ToN (follow-up journal):** Appropriate if the contribution is framed around graph-based QoS prediction and online map-healing in large-scale networks; INFOCOM has tracks on AI for networking and has seen REM and crowdsourcing work.<sup>[37][38][^39]</sup>
- **Workshops at NeurIPS/ICLR/KDD on Graph Learning for Communications or Spatio-Temporal Modeling:** A GNN-centric framing could be positioned for specialized workshops focusing on graph learning in physical networks, highlighting the physics-guided feature design.

### 4.3. Automotive / ITS venues

- **IEEE Transactions on Intelligent Transportation Systems (T-ITS):** Strong candidate for the ITS-focused angle, especially if the paper emphasizes route choice, vehicular QoS, and integration with predictive QoS standards for CAD and tele-operated driving.<sup>[3][24][34][1]</sup>
- **IEEE Intelligent Vehicles Symposium (IV) or IEEE ITSC:** Well-aligned conferences for presenting predictive QoS routing and vehicle-to-cloud map-healing as part of the connected and autonomous vehicle ecosystem.

In all cases, positioning the work as an integration of well-understood components (Okumura-Hata, GNNs, SegFormer, crowdsourced data) into a novel end-to-end architecture for predictive QoS routing will be key to convincing reviewers that the contribution is both practical and scientifically meaningful.

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